

```

import pandas as pd
import numpy as np
from datetime import datetime, timedelta

np.random.seed(42)
rows = 10500

data = {
    'Transaction_ID': range(1, rows + 1),
    'Date': [(datetime(2023, 1, 1) + timedelta(days=np.random.randint(0, 365)))],
    'Customer_Age': [np.random.randint(18, 70) if np.random.rand() > 0.1 else 18],
    'Product_Category': [np.random.choice(['Electronics', 'Clothing', 'Home', 'Food', 'Books', 'Toys', 'Garden', 'Pet Supplies', 'Sports', 'Travel', 'Holidays', 'Seasonal', 'Gifts', 'Miscellaneous'])],
    'Quantity': [np.random.randint(1, 10) if np.random.rand() > 0.05 else np.random.randint(1, 10)],
    'Unit_Price': [round(np.random.uniform(10, 500), 2) if np.random.rand() > 0.05 else round(np.random.uniform(10, 500), 2)],
    'Cost': [round(np.random.uniform(5, 300), 2) if np.random.rand() > 0.05 else round(np.random.uniform(5, 300), 2)]
}

df = pd.DataFrame(data)
df = pd.concat([df, df.iloc[:50]], ignore_index=True)
df.loc[df.sample(frac=0.01).index, 'Quantity'] = "Unknown"
df.to_csv('raw_dataset.csv', index=False)
print("Raw dataset created.")
print(df.head())

```

Raw dataset created.

	Transaction_ID	Date	Customer_Age	Product_Category	Quantity	Unit_Price	Cost
0			55.0	Home	9.0	101.93	31.42
1	Run this cell to mount your Google		50.0	Electronics	1.0	63.23	11.33
2	Drive.		60.0	Clothing	4.0	161.94	250.94
3	Learn more		62.0	Home	7.0	247.99	265.97
4			67.0	Clothing	7.0	401.20	207.79

Dismiss

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/tmp/ipykernel_1719/405291499.py:20: FutureWarning: Setting an item of in
df.loc[df.sample(frac=0.01).index, 'Quantity'] = "Unknown"

```

df = pd.read_csv('raw_dataset.csv')
print("Shape of raw data:", df.shape)
print("--- Missing Values BEFORE CLEANING ---")
print(df.isnull().sum())

```

```

Shape of raw data: (10550, 7)
--- Missing Values BEFORE CLEANING ---
Transaction_ID      0
Date                528
Customer_Age        1019
Product_Category    510

```

```
Quantity      484
Unit_Price     541
Cost           533
dtype: int64
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
df['Quantity'] = pd.to_numeric(df['Quantity'], errors='coerce')
df['Customer_Age'] = df['Customer_Age'].fillna(df['Customer_Age'].median())
df['Quantity'] = df['Quantity'].fillna(df['Quantity'].mode()[0])
df['Product_Category'] = df['Product_Category'].fillna('Unknown')
df.dropna(subset=['Unit_Price', 'Cost'], inplace=True)
Q1 = df['Quantity'].quantile(0.25)
Q3 = df['Quantity'].quantile(0.75)
IQR = Q3 - Q1
df = df[(df['Quantity'] >= Q1 - 1.5 * IQR) & (df['Quantity'] <= Q3 + 1.5 * IQR)]
df.drop_duplicates(inplace=True)
print("Data cleaning complete. New shape:", df.shape)
```

Data cleaning complete. New shape: (9450, 7)

```
df['Date'] = pd.to_datetime(df['Date'])
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
df['Total_Revenue'] = df['Unit_Price'] * df['Quantity']
df['Total_Cost'] = df['Cost'] * df['Quantity']
df['Profit'] = df['Total_Revenue'] - df['Total_Cost']
df['Profit_Margin_%'] = (df['Profit'] / df['Total_Revenue']) * 100
print(df.shape)
print(df.head())
```

Feature engineering complete.

	Transaction_ID	Date	Customer_Age	Product_Category	Quantity
0	1	2023-12-15	55.0	Home	9.0
1	2	2023-03-13	50.0	Electronics	1.0
2	3	2023-04-13	60.0	Clothing	4.0
3	4	2023-08-03	62.0	Home	7.0
4	5	2023-03-29	67.0	Clothing	7.0

	Unit_Price	Cost	Year	Month	Total_Revenue	Total_Cost	Profit
0	101.95	31.42	2023.0	12.0	917.55	282.78	634.77
1	63.23	11.33	2023.0	3.0	63.23	11.33	51.90
2	161.94	250.94	2023.0	4.0	647.76	1003.76	-356.00
3	247.99	265.97	2023.0	8.0	1735.93	1861.79	-125.86
4	401.20	207.79	2023.0	3.0	2808.40	1454.53	1353.87

	Profit_Margin_%
0	69.180971
1	82.081291
2	-54.958627
3	-7.250292
4	48.207876

```
df.to_csv('clean_dataset.csv', index=False)
print("Cleaned dataset saved successfully.")
```

Cleaned dataset saved successfully.

Run this cell to mount your Google Drive.

[Learn more](#)

